

E1.2 PRD - Dialogue-To-PRD Full Test

Date: 2026-06-26

Status: approved internal public-safe proof

Owner roles: AF Manager with AF Tools, AF Context, AF Knowledge, AF Review, and AF Publisher

Product

ArchFlow Block 1 Dialogue-to-PRD Knowledge Engine.

Problem

Strategic conversations and setup runs produce useful execution knowledge, but the material is easy to lose, repeat, or publish unsafely. ArchFlow needs a reproducible internal workflow that converts a sanitized source packet into PRD, task division, knowledge-base updates, review evidence, and a handout for the next agent.

Goal

Complete one full E1.2 proof package that demonstrates:

- public-safe source intake;
- approved-corpus retrieval;
- LangGraph-controlled parallel and sequential workflow stages;
- role-based CrewAI-style task division;
- PRD, streaming report, and system report generation;
- WikiLLM writeback;
- public-safety and runtime validation;
- Git commit and push readiness.

Users

User	Need
Founder/operator	Convert messy planning context into execution assets without repeating discovery.
Product manager	Get a PRD, task matrix, acceptance criteria, and risks from a controlled workflow.
Agent operator	Know which role owns each stage and how to continue without chat history.
Reviewer	Verify public-safety, evidence quality, runtime status, and unresolved gaps.

Non-Goals

- Do not ingest raw private transcripts, private Notion pages, private workspace links, screenshots, or credentials.
- Do not claim customer validation, ROI, or paying demand.
- Do not activate live Nexus writes.
- Do not promote Obsidian mirror to source of truth before the vault target is verified.
- Do not treat CrewAI LLM task execution as complete until it runs inside a LangGraph-reviewed proof.

Source Boundary

The E1.2 proof uses a sanitized source packet and public project files only.

Allowed	Blocked
project/, history/, skills/, wiki/, and graphify-out/GRAPH_REPORT.md	Raw private source material, secrets, private URLs, operational IDs, and local absolute paths

Workflow

```
flowchart TD
    START --> intake_validate
    intake_validate -->|pass| retrieve_context
    intake_validate -->|fail| review_gate
    retrieve_context --> analyze_architecture
    retrieve_context --> analyze_runtime
    retrieve_context --> analyze_knowledge
    retrieve_context --> analyze_agents
    retrieve_context --> analyze_trends
    analyze_architecture --> merge_findings
    analyze_runtime --> merge_findings
    analyze_knowledge --> merge_findings
    analyze_agents --> merge_findings
    analyze_trends --> merge_findings
    merge_findings --> draft_documents
    draft_documents --> review_gate
    review_gate -->|approved| publish_or_record
    review_gate -->|blocked| END
    publish_or_record --> END
```

Functional Requirements

ID	Requirement	Owner	Acceptance
FR1	Validate sanitized source boundary.	AF Tools	Boundary status is pass or output is blocked.
FR2	Retrieve approved public context.	AF Context	Source paths are present and private-source refusal remains active.
FR3	Run parallel expert analysis.	LangGraph plus role agents	Architecture, runtime, knowledge, agent, and trends branches complete before merge.

ID	Requirement	Owner	Acceptance
FR4	Produce the PRD.	AF Manager	PRD includes goal, scope, workflow, requirements, task division, risks, and acceptance.
FR5	Produce streaming report.	AF Tools plus AF Review	Report stores observable stream updates, not hidden chain-of-thought.
FR6	Produce system report.	AF Architecture plus AF Knowledge	Report reviews runtime, roles, models, temperature, efficiency, KB, Graphify, Nexus, and vault posture.
FR7	Produce task matrix and KB update.	AF Manager and AF Knowledge	E1.2 outputs define E1.3 next steps and durable memory candidates.
FR8	Run publication gate.	AF Review and AF Publisher	Public safety scan and runtime guard pass before commit/push.

LangGraph State

Field	Purpose
run_id	Stable run identifier.
source_packet_id	Sanitized source reference.
source_boundary_status	pass or fail.
retrieved_context	Approved source-path grounded context.
architecture_analysis	Control-path and layer design findings.
runtime_analysis	Package, smoke, trace, and guard findings.
knowledge_analysis	WikiLLM, Graphify, Obsidian, and Nexus findings.
agent_analysis	Role, model, temperature, and accountability findings.
trends_analysis	Official-doc tendencies and recommended modern practices.
merged_findings	Integrated conclusions from parallel branches.
approval_status	approved, revise, or blocked.
stream_notes	Public-safe node progress summaries.

Agentic Role Division

Agent	Role	Model/Temperature Policy	Current Status
AF Tools	Source, runtime, and tool readiness.	Deterministic, temperature 0.0-0.2.	Used for validation and graph evidence.
AF Context	Source digest and FACT/INTERPRETATION/HYPOTHESIS/GAP split.	Low variance, temperature 0.2-0.4.	Used for digest and retrieval framing.
AF Manager	PRD, task division, milestones, and acceptance criteria.	Low-to-moderate, temperature 0.2-0.5.	Used for PRD and matrix.

Agent	Role	Model/Temperature Policy	Current Status
AF Knowledge	KB update, memory, insight, issue, and decision candidates.	Low variance, temperature 0.1-0.3.	Used for WikiLLM outputs.
AF Review	Safety, evidence, unsupported-claim, and publication gate.	Strict deterministic, temperature 0.0-0.1.	Used for approval criteria.
AF Publisher	Git-ready report pack and handout.	Deterministic, temperature 0.0.	Used for final packaging.
Technical Tendencies Analyst	Official-doc trend check and config recommendations.	Moderate, temperature 0.3-0.5.	Used for latest-approach review.

Task Division

Task	Owner	Dependency	Output
E1.2.1 Source boundary and inventory	AF Tools	E1.1 runtime ready	source-inventory.md
E1.2.2 Retrieval and context digest	AF Context	E1.2.1	retrieval-log.md, context-digest.md
E1.2.3 Full-test graph run	LangGraph controller	E1.2.2	e1_2_langgraph_output.json, e1_2_stream_events.jsonl
E1.2.4 PRD and role plan	AF Manager	E1.2.3	E1.2_PRD.md, responsibility-matrix.md
E1.2.5 Streaming report	AF Tools and AF Review	E1.2.3	E1.2_Streaming_report.md
E1.2.6 System report	AF Architecture and AF Knowledge	E1.2.3	E1.2_System_report.md
E1.2.7 Review and KB writeback	AF Review and AF Knowledge	E1.2.4-E1.2.6	review-report.md, kb-update.md, WikiLLM run note
E1.2.8 PDF render, validate, commit, push	AF Publisher	E1.2.7	Three PDFs, commit, remote push

Acceptance Criteria

- E1.2 full-test graph runs and reaches approved status.
- The graph saves stream events and final state.
- PRD, streaming report, system report, task matrix, KB update, review report, run summary, and handout exist.
- The three requested PDFs exist and are text-extractable.
- Public files do not contain secrets, private URLs, local absolute paths, or unmasked operational IDs.
- Workflow validation, LangGraph smoke, LlamaIndex retrieval, CrewAI config check, runtime guard, dashboard regeneration, and public safety scan pass.
- Git commit and push complete after validation.

Risks

Risk	Mitigation
Overstating E1.2 as customer validation.	Label it internal proof only.
Saving hidden reasoning as a report.	Store observable stream events and state summaries only.
CrewAI treated as full execution proof too early.	Keep CrewAI LLM execution behind LangGraph review.
Nexus activated before vault schemas are known.	Keep Nexus planned until schema discovery and vault readiness pass.
Dashboard data stales after edits.	Regenerate dashboard data before final validation.

Latest Technical Tendencies Checked

Official docs checked on 2026-06-26 support this direction: LangGraph for durable agent workflows, persistence, streaming, and human-in-the-loop control; CrewAI for crews/processes and role/task accountability; LlamaIndex for agentic workflows and RAG over data; and LangSmith for tracing and observability. Sources are listed in the system report.

Next Execution

E1.3 should write the approved PRD and agent history into the KB, then run a readback test proving another agent can answer from the public memory without re-reading this chat.